

Homework 3

MATH 220

Problem 1 (Examples and nonexamples of bases)

- i) Prove that $\mathcal{B} = \{]p, q[\mid p < q \in \mathbb{Q}\}$ is a basis for the Euclidean topology on $\mathbb{R}$.
- ii) Prove that $\mathcal{B}' = \{[p, q[\mid p < q \in \mathbb{Q}\}$ is not a basis for the lower limit topology on $\mathbb{R}$.
- iii) Is $\mathcal{B}'$ from the previous point the basis for a topology on $\mathbb{R}$? Prove your claim.
- iv) Is the family $\mathcal{B}'' = \{[a, b] \mid a, b \in \mathbb{R}, a < b\}$ the basis for a topology on $\mathbb{R}$? Prove your claim.

Problem 2 (The K -topology on $\mathbb{R}$)

Let

$$\mathcal{B}_K = \{]a, b[\mid a, b \in \mathbb{R}, a < b\} \cup \{]a, b[\setminus K \mid a, b \in \mathbb{R}, a < b\}$$

where $K = \{n^{-1} \mid n \in \mathbb{N}\}$.

- i) Prove that $\mathcal{B}_K$ is the basis for a topology on $\mathbb{R}$. We will call this topology the K -topology and denote it by $\mathcal{T}_K$.
- ii) Prove that $\mathcal{T}_K$ is strictly finer than the Euclidean topology.
- iii) Prove that $\mathcal{T}_K$ and the lower limit topology are not comparable.

Problem 3 (A set is open iff it is a neighborhood of all its points)

Let X be a topological space and $A \subseteq X$. Prove that A is open if and only if $A \in \mathcal{N}(x)$ for all $x \in A$.

Problem 4 (Examples of interiors of sets)

- i) It can be shown that $\mathcal{B}_R = \{]a, b[\mid a, b \in \mathbb{R}, a < b\}$ is the basis for a topology on $\mathbb{R}$ called the upper limit topology. In this topology, find the interior of the sets $A =]a, b[$ and $B = [a, b]$, where $a, b \in \mathbb{R}, a < b$. Prove your claims.
- ii) In $\mathbb{R}$ equipped with the cofinite topology, find the interior of the following sets: $A =]a, b[$, $B =]a, \infty[$, $C =]-\infty, a[\cup]a, \infty[$, where $a, b \in \mathbb{R}, a < b$. Prove your claims.

The next problem is optional (for fun) and will not be graded. It is an upgraded version of the intuitive discussion we had in class about families of discs and squares in the plane.

Problem * For $x \in \mathbb{R}^n$ and $r > 0$ define

$$D_r(x) = \{y \in \mathbb{R}^n \mid \sum_{i=1}^n (y_i - x_i)^2 < r^2\},$$

$$Q_r(x) = \{y \in \mathbb{R}^n \mid \max_{i=1, \dots, n} |y_i - x_i| < r\},$$

and

$$R_r(x) = \{y \in \mathbb{R}^n \mid \sum_{i=1}^n |y_i - x_i| < r\},$$

Consider the following collections of sets

$$\mathcal{B} = \{D_r(x) \mid x \in \mathbb{R}^n, r > 0\},$$

$$\mathcal{Q} = \{Q_r(x) \mid x \in \mathbb{R}^n, r > 0\}$$

and

$$\mathcal{R} = \{R_r(x) \mid x \in \mathbb{R}^n, r > 0\}.$$

Prove that $\mathcal{B}$, $\mathcal{Q}$, and $\mathcal{R}$ are bases for the Euclidean topology on $\mathbb{R}^n$.